```sql
WITH CitySales AS (
    SELECT L.Country, L.City, SUM(F.Sales) AS TotalSales
    FROM Fact F
    JOIN Location L ON F.[Location ID] = L.[Location ID]
    GROUP BY L.Country, L.City
)
SELECT Country, City, TotalSales
FROM (
    SELECT
      Country,
      City,
      TotalSales,
      DENSE_RANK() OVER(PARTITION BY Country ORDER BY TotalSales DESC) AS
      RowNum
    FROM CitySales
) AS RankedCities
WHERE RowNum <= 3
```

Country	City	TotalSales
Afghanistan	Kabul	16951.11
Afghanistan	Kandahar	1867.02
Afghanistan	Herat	1695.24
Albania	Elbasan	1619.1
Albania	Vlore	986.49
Albania	Durres	612.09
Algeria	Algiers	9270.06
Algeria	Oran	7337.46
Algeria	Guelma	3664.71
.	.	.
.	.	.
.	.	.
Yemen	Taizz	2032.173
Yemen	Ibb	354.564
Yemen	Al Hudaydah	78.975
Zambia	Lusaka	12767.22
Zambia	Kitwe	3177.66
Zambia	Ndola	2893.8
Zimbabwe	Bulawayo	2080.233
Zimbabwe	Harare	785.7
Zimbabwe	Chitungwiza	726.462

```sql
WITH YearlySales AS (
    SELECT
        YEAR(F.[Order Date]) AS SalesYear,
        SUM(F.Sales) AS TotalSales
    FROM Fact F
    GROUP BY YEAR(F.[Order Date])
),
Growth AS (
    SELECT
        SalesYear,
        TotalSales,
        LAG(TotalSales) OVER (ORDER BY SalesYear) AS PreviousYearSales
    FROM YearlySales
)
SELECT
    SalesYear,
    TotalSales,
    (TotalSales - PreviousYearSales) * 100.0 / PreviousYearSales AS
YoYGrowthPercentage
FROM Growth
```

SalesYear	TotalSales	YoYGrowthPercentage
2020	19843.05704	NULL
2021	2265592.20439999	11317.5562758952
2022	2700389.87903998	19.1913475777135
2023	3391894.65767997	25.6075903708331
2024	4264782.11171998	25.7345095332961

```sql
SELECT L.Country, L.City, AVG(F.Sales) AS AvgCitySales
FROM Fact F
JOIN Location L ON F.[Location ID] = L.[Location ID]
GROUP BY L.Country, L.City
```

	Country	City	AvgCitySales
1	Germany	Aachen	223.714764705882
2	Germany	Aalen	979.695
3	Belgium	Aalst	96.8325
4	Nigeria	Aba	42.56496
5	Iran	Abadan	123.177272727273
.	.	.	.
.	.	.	.
.	.	.	.
3699	China	Zunyi	312
3700	Switzerland	Zurich	368.547931034483
3701	Liberia	Zwedru	49.41
3702	Germany	Zwickau	33.01
3703	Netherlands	Zwolle	75.2925

```sql
WITH CitySales AS (
    SELECT
        L.Country,
        L.City,
        SUM(F.Sales) AS TotalSales
    FROM Fact F
    JOIN Location L ON F.[Location ID] = L.[Location ID]
    GROUP BY L.Country, L.City
),
CountryCityCount AS (
    SELECT
        Country,
        COUNT(City) AS CitiesWithSalesAbove10000
    FROM CitySales
    WHERE TotalSales > 10000
    GROUP BY Country
)
SELECT COUNT(Country) AS CountryCount
FROM CountryCityCount
WHERE CitiesWithSalesAbove10000 >= 5
```

	CountryCount
1	14

```sql
SELECT F.[Customer ID], AVG(F.Sales) AS AvgOrderValue
FROM Fact F
GROUP BY F.[Customer ID]
ORDER BY AvgOrderValue
```

	Customer ID	AvgOrderValue
1	ZC-11910	7.173
2	BD-1500	17.349
3	MG-7890	19.128
4	MP-7470	19.37
5	SH-10395	21.505
.	.	.
.	.	.
.	.	.
1586	HL-15040	631.15384212766
1587	MG-8145	653.63925
1588	DJ-3510	664.076666666667
1589	SM-20320	798.084486153846
1590	BW-1065	902.748

```sql
SELECT
COUNT(DISTINCT F.[Customer ID]) AS IndianCustomersWithLoss
FROM Fact F
JOIN Customer C ON F.[Customer ID] = C.[Customer ID]
JOIN Location L ON F.[Location ID] = L.[Location ID]
WHERE L.Country = 'India'
AND F.Profit < 0
```

	IndianCustomersWithLoss
1	35

```sql
SELECT C.Segment, P.[Sub-Category], SUM(F.Sales) AS TotalSales
FROM Fact F
JOIN Product P ON F.[Product ID] = P.[Product ID]
JOIN Customer C ON F.[Customer ID] = C.[Customer ID]
GROUP BY C.Segment, P.[Sub-Category]
ORDER BY C.Segment, P.[Sub-Category], TotalSales
```

	Segment	Sub-Category	TotalSales
1	Consumer	Accessories	382163.2654
2	Consumer	Appliances	510230.2822
3	Consumer	Art	200870.6144
4	Consumer	Binders	253750.7256
5	Consumer	Bookcases	765111.1448
6	Consumer	Chairs	778362.5301
7	Consumer	Copiers	757081.41666
8	Consumer	Envelopes	88473.9481
9	Consumer	Fasteners	42074.4092
10	Consumer	Furnishings	203195.115
11	Consumer	Labels	38692.4192
12	Consumer	Machines	382373.0451
13	Consumer	Paper	119654.7791
14	Consumer	Phones	905422.2842
15	Consumer	Storage	575506.1567
16	Consumer	Supplies	123260.3051
17	Consumer	Tables	381726.977
18	Corporate	Accessories	231023.6969
19	Corporate	Appliances	317655.4142
20	Corporate	Art	103571.3772
21	Corporate	Binders	128548.3657
22	Corporate	Bookcases	457327.1684
23	Corporate	Chairs	448519.5128
24	Corporate	Copiers	462774.31806
25	Corporate	Envelopes	51805.8048
26	Corporate	Fasteners	26138.4157
27	Corporate	Furnishings	112879.404
28	Corporate	Labels	21178.6937
29	Corporate	Machines	229907.7571
30	Corporate	Paper	71982.5453
31	Corporate	Phones	494085.5843
32	Corporate	Storage	340019.3634
33	Corporate	Supplies	81486.3971
34	Corporate	Tables	245793.7028
35	Home Office	Accessories	136050.0562
36	Home Office	Appliances	183178.6086
37	Home Office	Art	67649.9743
38	Home Office	Binders	79632.0704
39	Home Office	Bookcases	244133.9286
40	Home Office	Chairs	274799.7213
41	Home Office	Copiers	289580.53856
42	Home Office	Envelopes	30624.5487
43	Home Office	Fasteners	15029.491
44	Home Office	Furnishings	69503.7369
45	Home Office	Labels	13513.2611
46	Home Office	Machines	166779.2649
47	Home Office	Paper	52654.395
48	Home Office	Phones	307316.2707
49	Home Office	Storage	211560.3413
50	Home Office	Supplies	38327.5184
51	Home Office	Tables	129521.2446

```sql
WITH QuarterlySales AS (
    SELECT
        YEAR(F.[Order Date]) AS SalesYear,
        DATEPART(QUARTER, F.[Order Date]) AS SalesQuarter,
        SUM(F.Sales) AS TotalSales
    FROM Fact F
    JOIN Location L ON F.[Location ID] = L.[Location ID]
    WHERE L.Country = 'India'
    GROUP BY YEAR(F.[Order Date]), DATEPART(QUARTER, F.[Order Date])
),
QuarterlyGrowth AS (
    SELECT
        SalesYear,
        SalesQuarter,
        TotalSales,
        LAG(TotalSales) OVER (ORDER BY SalesYear, SalesQuarter) AS
  PreviousQuarterSales
    FROM QuarterlySales
)
SELECT
    SalesYear,
    SalesQuarter,
    TotalSales,
    (TotalSales - PreviousQuarterSales) * 100.0 / PreviousQuarterSales AS
QoQGrowthPercentage
FROM QuarterlyGrowth
```

	SalesYear	SalesQuarter	TotalSales	QoQGrowthPercentage
1	2021	2	13546.065	56.9553746471823
2	2021	3	30921.435	128.26876292118
3	2021	4	40385.205	30.605856422899
4	2022	1	29105.4	-27.930537928432
5	2022	2	35263.56	21.1581356037024
6	2022	3	34079.16	-3.35870796935986
7	2022	4	41571.225	21.9843006693827
8	2023	1	21721.635	-47.7483884586033
9	2023	2	44316.54	104.020277479112
10	2023	3	52045.515	17.440384560708
11	2023	4	33692.49	-35.2634131874764
12	2024	1	32840.01	-2.53017809013226
13	2024	2	45230.58	37.7301042234762
14	2024	3	63124.215	39.5609231630459
15	2024	4	63176.55	0.0829079617069284

```sql
WITH cte AS(
    SELECT P.[Product Name], SUM(F.Sales) [Total Sales], DENSE_RANK()
OVER(ORDER BY SUM(F.Sales) DESC) rk
    FROM Fact F JOIN Product P ON F.[Product ID] = P.[Product ID]
    WHERE DATEPART(QUARTER, [Order Date]) = DATEPART(QUARTER, GETDATE())
    GROUP BY P.[Product Name]
)
SELECT [Product Name], [Total Sales]
FROM cte
WHERE rk <=10
ORDER BY [Total Sales] DESC
```

	Product Name	*Total Sales*
1	Cisco TelePresence System EX90 Videoconferencing Unit	22638.48
2	Motorola Smart Phone, Full Size	15602.196
3	SAFCO Executive Leather Armchair, Black	15215.85
4	Canon imageCLASS 2200 Advanced Copier	13999.96
5	KitchenAid Refrigerator, Black	11419.2
6	Barricks Conference Table, Fully Assembled	10902.6
7	Sauder Classic Bookcase, Traditional	10594.557
8	Hon Executive Leather Armchair, Adjustable	10225.451
9	Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	10167.92
10	Martin Yale Chadless Opener Electric Letter Opener	9660.596

```sql
SELECT
    C.Segment,
    AVG(F.Profit) AS AverageProfitPerOrder
FROM
    Fact F
    JOIN Customer C ON F.[Customer ID] = C.[Customer ID]
GROUP BY
    C.Segment
ORDER BY
    C.Segment
```

	Segment	*AverageProfitPerOrder*
1	Consumer	28.2540079214119
2	Corporate	28.5960417823579
3	Home Office	29.6488473252702

```sql
SELECT
    P.[Sub-Category],
    SUM(F.Profit) AS TotalProfit,
    (SUM(F.Profit) * 100.0 / (SELECT SUM(Profit) FROM Fact)) AS PercentProfit
FROM
    Fact F
    JOIN Product P ON F.[Product ID] = P.[Product ID]
GROUP BY
    P.[Sub-Category]
ORDER BY
    PercentProfit DESC
```

	Sub-Category	*TotalProfit*	*PercentProfit*
1	Copiers	258567.54818	17.6201072233225
2	Phones	216717.0058	14.7681985082487
3	Bookcases	161924.4195	11.0343531264723
4	Appliances	141680.5894	9.6548356290777
5	Chairs	140396.2675	9.56731540565234
6	Accessories	129626.3062	8.83339549098104
7	Storage	108461.4898	7.39111730504904
8	Binders	72451.502	4.93721367092078
9	Paper	59207.6827	4.03471249567718
10	Machines	58867.873	4.01155613521481
11	Art	57953.9109	3.9492741113746
12	Furnishings	46967.4255	3.20059914377694
13	Envelopes	29601.1163	2.01717054907814
14	Supplies	22583.2631	1.53893835508505
15	Labels	15008.856	1.0227797489703
16	Fasteners	11525.4241	0.785400990440196
17	Tables	-64083.3887	-4.36696788934162

```sql
WITH TopProfitableCustomers AS (
    SELECT TOP 5
        C.[Customer ID],
        C.[Customer Name],
        SUM(F.Profit) AS TotalProfit
    FROM
        Fact F
        JOIN Customer C ON F.[Customer ID] = C.[Customer ID]
    GROUP BY
        C.[Customer ID], C.[Customer Name]
    ORDER BY
        TotalProfit DESC
),
TopLossCustomers AS (
    SELECT TOP 5
        C.[Customer ID],
        C.[Customer Name],
        SUM(F.Profit) AS TotalProfit
    FROM
        Fact F
        JOIN Customer C ON F.[Customer ID] = C.[Customer ID]
    GROUP BY
        C.[Customer ID], C.[Customer Name]
    ORDER BY
        TotalProfit ASC
)
SELECT * FROM TopProfitableCustomers
UNION ALL
SELECT * FROM TopLossCustomers
```

	Customer ID	Customer Name	TotalProfit
1	TC-20980	Tamara Chand	8787.4749
2	RB-19360	Raymond Buch	8523.9515
3	SC-20095	Sanjit Chand	8106.2179
4	BE-11335	Bill Eplett	7790.6963
5	HL-15040	Hunter Lopez	7657.50178
6	CS-12505	Cindy Stewart	-6437.3681
7	DM-3345	Denise Monton	-5474.61
8	GT-14635	Grant Thornton	-3790.08306
9	LF-17185	Luke Foster	-3700.1955
10	MT-8070	Michelle Tran	-2991.618

```sql
SELECT TOP 1
    p.Category,
    c.Segment,
    SUM(f.Profit) / (SELECT SUM(Profit) FROM Fact) * 100 AS PercentProfit
FROM
    Fact f
    LEFT JOIN Product p ON f.[Product ID] = p.[Product ID]
    LEFT JOIN Customer c ON f.[Customer ID] = c.[Customer ID]
GROUP BY
    p.Category, c.Segment
ORDER BY
    PercentProfit DESC
```

Category	Segment	PercentProfit
Technology	Consumer	23.3359707294309

```sql
SELECT DISTINCT [Customer ID]
FROM Fact
WHERE [Customer ID] NOT IN (
    SELECT DISTINCT [Customer ID]
    FROM Fact
    WHERE [Order Date] > GETDATE() - 100
)
```

	Customer ID	*AvgOrderValue*
1	KL-6645	105.0864
2	EH-3945	132.565263157895
3	HA-4905	110.6604
4	KC-6675	97.9605
5	BW-11110	377.945882716049
.	.	
.	.	
.	.	
1586	PO-9180	58.6222857142857
1587	AW-10840	220.004258666667
1588	DL-12865	203.376692727273
1589	KD-16270	374.262379032258
1590	AS-10135	208.778405714286

```sql
SELECT F.[Customer ID]
FROM Fact F
GROUP BY F.[Customer ID]
HAVING COUNT(F.[Order ID]) = 1
```

	Customer ID
1	MG-7650
2	ME-8010
3	MG-7890
4	BG-1035
5	RC-9825
6	ZC-11910
7	DK-2985

```sql
WITH ProductSales AS (
    SELECT
        L.State,
        P.[Product Name],
        DENSE_RANK() OVER (PARTITION BY State ORDER BY SUM(F.Quantity) DESC)
AS ProductRank
    FROM
        Fact F
        JOIN Product P ON F.[Product ID] = P.[Product ID]
        JOIN Location L ON F.[Location ID] = L.[Location ID]
    WHERE
        L.Country = 'India'
    GROUP BY
        L.State, P.[Product Name]
)
SELECT State, [Product Name]
FROM ProductSales
WHERE ProductRank = 1
```

	State	Product Name
1	Andhra Pradesh	Rubbermaid Photo Frame, Black
2	Assam	Enermax Memory Card, USB
3	Bihar	Smead File Cart, Industrial
4	Chandigarh	Cardinal Index Tab, Economy
5	Chhattisgarh	Nokia Office Telephone, Full Size
.	.	.
.	.	.
.	.	.
26	Telangana	Accos Clamps, Bulk Pack
27	Tripura	Eaton Memo Slips, Multicolor
28	Uttar Pradesh	Novimex Chairmat, Set of Two
29	Uttarakhand	Safco Classic Bookcase, Pine
30	West Bengal	Smead Shipping Labels, Alphabetical

```sql
SELECT
 [Customer ID],
    MIN([Order Date]) AS FirstOrderDate,
    MAX([Order Date]) AS LastOrderDate,
    DATEDIFF(DAY, MIN([Order Date]), MAX([Order Date])) AS DaysBetween
FROM
 Fact
GROUP BY
 [Customer ID]
```

	Customer ID	FirstOrderDate	LastOrderDate	DaysBetween
1	KL-6645	2023-06-05 00:00:00.000	2024-12-24 00:00:00.000	568
2	HA-4905	2023-06-12 00:00:00.000	2024-12-11 00:00:00.000	548
3	JE-16165	2021-10-12 00:00:00.000	2024-12-04 00:00:00.000	1149
4	SJ-20125	2021-12-05 00:00:00.000	2024-11-22 00:00:00.000	1083
5	DK-3150	2021-01-25 00:00:00.000	2024-10-28 00:00:00.000	1372
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
1586	JR-5670	2022-03-24 00:00:00.000	2024-08-18 00:00:00.000	878
1587	PO-9180	2021-02-03 00:00:00.000	2024-11-25 00:00:00.000	1391
1588	DL-12865	2021-01-03 00:00:00.000	2024-10-28 00:00:00.000	1394
1589	KD-16270	2021-09-13 00:00:00.000	2024-11-18 00:00:00.000	1162
1590	AS-10135	2021-11-10 00:00:00.000	2024-11-13 00:00:00.000	1099

```sql
SELECT YEAR(F.[Order Date]) AS Year,
       MONTH(F.[Order Date]) AS Month,
       SUM(F.Sales) AS TotalSales
FROM Fact F
WHERE F.[Order Date] >= DATEDIFF(MONTH, 10, GETDATE())
GROUP BY YEAR(F.[Order Date]), MONTH(F.[Order Date])
ORDER BY Year, Month ASC
```

	Year	Month	TotalSales
1	2020	12	19843.05704
2	2021	1	93480.47292
3	2021	2	108996.87258
4	2021	3	149945.48544
5	2021	4	113423.49568
6	2021	5	179180.87298
7	2021	6	168349.92614
8	2021	7	140168.13834
9	2021	8	207563.39128
10	2021	9	297139.862620001
11	2021	10	239397.30476
12	2021	11	272172.8103
13	2021	12	295773.57136
14	2022	1	142164.51748
15	2022	2	97575.2933600001
16	2022	3	160787.48464
17	2022	4	158927.73618
18	2022	5	226190.31664
19	2022	6	248227.68502
20	2022	7	172056.28556
21	2022	8	295660.08428
22	2022	9	296514.30842
23	2022	10	246085.3665
24	2022	11	343255.40626
25	2022	12	312945.3947
26	2023	1	194575.87802
27	2023	2	168531.81606
28	2023	3	190054.09638
29	2023	4	180587.96144
30	2023	5	290946.48434
31	2023	6	373013.02606
32	2023	7	235526.5987
33	2023	8	355849.538640001
34	2023	9	363486.679140001
35	2023	10	294812.7728
36	2023	11	385725.16026
37	2023	12	358784.64584
38	2024	1	233153.99422
39	2024	2	196807.04584
40	2024	3	277194.533460001
41	2024	4	241235.00528
42	2024	5	319863.451260001
43	2024	6	383083.18334
44	2024	7	306594.23454
45	2024	8	458446.43302
46	2024	9	469574.0791
47	2024	10	434286.525220001
48	2024	11	552622.415980001
49	2024	12	391921.21046

```sql
SELECT P.[Product Name], AVG(F.Sales / F.Quantity) AS AvgSellingPrice
FROM Fact F
JOIN Product P ON F.[Product ID] = P.[Product ID]
GROUP BY P.[Product Name]
ORDER BY AvgSellingPrice DESC
```

	Product Name	*AvgSellingPrice*
1	Cisco TelePresence System EX90 Videoconferencing Unit	3773.08
2	Canon imageCLASS 2200 Advanced Copier	3079.9912
3	Cubify CubeX 3D Printer Triple Head Print	1999.995
4	Canon imageCLASS MF7460 Monochrome Digital Laser Multifunction Copier	1995.99
5	High Speed Automatic Electric Letter Opener	1528.36133333333
.	.	.
.	.	.
.	.	.
3656	Avery Triangle Shaped Sheet Lifters, Black, 2/Pack	0.943
3657	Maxell 4.7GB DVD+R 5/Pack	0.9108
3658	Maxell 4.7GB DVD-R 5/Pack	0.9108
3659	Avery Reinforcements for Hole-Punch Pages	0.8415
3660	Eureka Disposable Bags for Sanitaire Vibra Groomer I Upright Vac	0.812

```sql
SELECT
    L.Country,
    AVG(DATEDIFF(DAY, F.[Order Date], F.[Ship Date])) AS AverageShippingDays
FROM
    Fact F
    JOIN Location L ON F.[Location ID] = L.[Location ID]
GROUP BY
    L.Country
ORDER BY
    AverageShippingDays ASC
```

	Country	AverageShippingDays
1	Bahrain	2
2	Liberia	2
3	Guinea-Bissau	2
4	Namibia	2
5	Guadeloupe	2
143	Tajikistan	5
144	Armenia	5
145	Jamaica	5
146	Macedonia	5
147	South Sudan	6